# Stochastic Evolutionary Game Theory

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## Abstract

This work analyzes evolutionary strategies in finite populations through stochastic birth-death dynamics, using the Moran process and the Fermi function. We calculate the fixation probability and mean fixation time of a mutant, connecting this dynamics with the replicator equation. We study both the stochastic and deterministic regimes, identifying the four classical evolutionary scenarios. The analysis extends to fluctuating environments, exploring the dilemma between specialized and generalist strategies, showing that the rate of environmental change can favor riskier behaviors. Finally, we apply this framework to tumors, demonstrating how cell interactions guide evolution and suggesting therapies that modify the environment rather than directly attacking cancer cells.

## 1 Introduction

In the early twentieth century, John von Neumann developed a mathematical branch in which he intended to analyze strategic decisions in contexts of interaction among multiple rational agents. Von Neumann defined classical game theory as "a theory dealing with the determination of optimal strategies in competitive games, where the outcome for each player depends on the choices of all participants" (Von Neumann and Morgenstern, 1944).

Evolutionary game theory, on the other hand, advocates for the study of strategies that lead to greater success in interaction within a population and tend to prevail, either through reproductive capacity or by imitation of successful behaviors, without assuming rationality or conscious decision-making: "evolutionary game theory is the application of game theory to populations in evolution within biology. It defines a framework of competitions, strategies, and analytical tools through which Darwinian competition can be modeled" (Maynard Smith, 1982).

The cornerstone of interactions between agents is the payoff matrix, which defines the benefit that each agent receives when facing different combinations of strategies:

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

The value $a$ represents the payoff obtained by a player using strategy A when facing another player who also uses A; $b$ is the payoff of A when playing against a player using strategy B; $c$ is the payoff of B when facing a player with strategy A; and finally, $d$ is the payoff received by a player with strategy B when playing against another player who also uses B (Traulsen and Hauert, 2009).

To analyze how strategies evolve within a population, both replicator dynamics and stochastic dynamics are used. To relate both, it is fundamental to consider the large population limit to recover deterministic dynamics.

# 2 Stochastic Evolutionary Dynamics

We consider a system of $N$ individuals interacting among themselves. Suppose $j$ of them adopt strategy A and the remaining $N-j$ adopt strategy B. The probability of randomly selecting a pair consisting of one individual with strategy A and another with strategy B is $\frac{j(N-j)}{N^2}$.

To model the probability that an individual adopts the strategy of another, a stochastic rule based on the Fermi distribution is employed, which depends on the difference in payoffs $\Delta\pi = \pi_A - \pi_B$ between strategies. The total payoff received by an individual with strategy A and another with strategy B is respectively:

$$\pi_A(j) = \frac{j-1}{N}a + \frac{N-j}{N}b \quad \text{and} \quad \pi_B(j) = \frac{j}{N}c + \frac{N-j-1}{N}d$$

The transition probabilities are:

- Probability that an individual with strategy B adopts strategy A (i.e., $j \to j+1$): $T_j^+ = \frac{j(N-j)}{N^2} \cdot \frac{1}{1+\exp(-w\Delta\pi)}$
- Probability that an individual with strategy A adopts strategy B (i.e., $j \to j-1$): $T_j^- = \frac{j(N-j)}{N^2} \cdot \frac{1}{1+\exp(w\Delta\pi)}$

## 2.1 Fixation Probability

The fixation probability of a mutant is:

$$\phi_1 = \frac{1}{1 + \sum_{m=1}^{N-1} \prod_{j=1}^{m} \xi_j}$$

where $\xi_j = \frac{T_j^-}{T_j^+}$.

## 2.2 Fixation Time

The fixation time of a mutant is:

$$t_1 = \phi_1 \left( \sum_{w=1}^{N-1} \sum_{l=1}^{w} \frac{1}{T_l^+} \prod_{m=l+1}^{w} \xi_m \right)$$

# 3 Fixation in Fluctuating Environments

We consider a mutant of type A introduced into a finite population of $N-1$ residents of type B. Environmental switching between two states ($\sigma = \pm 1$) with rates $p_+$ and $p_-$ introduces an additional source of stochasticity.

The payoff matrix becomes:

$$M_\sigma = \begin{bmatrix} 1 & 1+0.5\sigma \\ 1+0.9\sigma & 1 \end{bmatrix}$$

The analysis reveals that both too slow and too rapid environmental alternation impair fixation, explaining why a maximum in fixation probability is observed around $p_+ \approx p_-$.

# 4 Tumor Dynamics

Tumors represent a direct application of game theory, as cancer cells have lost their normal cooperative behavior and defect. However, results indicate that tumors do not consist entirely of cancer cells, suggesting that tumors may not be the result of random mixing but rather an equilibrium due to interactions between tumor and normal cells (Gerstung, Nakhoul, and Beerenwinkel, 2011).

In their study on multiple myeloma (MM), a type of bone marrow cancer, Pacheco, Santos, and Dingli (2014) analyze, from an evolutionary game theory perspective, the equilibrium of bone remodeling, determined by the interaction between bone resorption—carried out by osteoclasts (OC)—and bone formation—performed by osteoblasts (OB).

From a medical perspective, therapies that reduce $\beta$ decrease lesions and slow cancer progression, while drugs that reduce $\delta$ improve bone mass and reduce the number of cancer cells.

## 5 Conclusions

This work has explored how evolutionary game theory can serve as a rigorous framework for connecting mathematical models with observable biological phenomena. The equations developed are not merely instruments of quantitative analysis but formalize evolutionary principles relevant in different fields of biology.

The analysis of fluctuating environments has identified a non-trivial evolutionary dynamic: optimal adaptability is not achieved under conditions of complete stability nor extreme variability, but in intermediate regimes.

The framework developed offers relevant implications in different research areas. In oncology, it suggests therapeutic approaches based on modifying the tumor microenvironment rather than directly intervening on cancer cells. In ecology, it provides tools for modeling the viability of invasive species under non-stationary environmental conditions.

## References


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